Alexander King, PhD

Applied AI Researcher & Computational Scientist

Riverside, CA | Open to remote and California-based roles

alexander.julian.king@gmail.com
(925) 451-6310
alexanderjulianking.github.io
linkedin.com/in/alexander-julian-king
github.com/AlexanderJulianKing

Profile

Applied AI researcher and computational scientist who develops original methods, validates them against controlled baselines, and ships working systems. Neuroscience PhD and first-author in the *Journal of Neuroscience*; public work spans agent failure analysis, LLM benchmarking, scientific-audit software, and agentic research automation.

AI Evaluation & Reliability

Silent Precondition Traps in Data-Science Agents — 2026

Independent study; public manuscript and permanently archived artifacts — Zenodo | GitHub

- Designed a 1,000-run study using two frozen task batteries to test whether data-science agents catch hidden precondition failures. Agents reproduced the exact planted error in 329 of 330 incorrect answers, while a short precondition-checking prompt cut errors roughly fourfold.

SC-Referee — 2026

Selected for Anthropic's Built with Claude: Life Sciences hackathon (Builder track) — GitHub

- Built SC-Referee, a deterministic scientific-analysis checker delivered as a Python application and Claude Code skill. It reconstructs analysis designs from heterogeneous data, results, and code, then requires scientist confirmation before running 12+ verifiers. A published melanoma reanalysis reduced the significant-gene count from **16,289 to 770** at the biological-sample level.

Applied AI Systems

Project Soothsayer — Solo-built

LLM evaluation & score-prediction system — GitHub

- Built an Arena.ai human-preference predictor from 138 labeled models, 122 features, and roughly 40% missing data using custom imputation and adaptive local regression. Soothsayer reduced RMSE 29% versus the best regularized baseline and achieved $\mathbf{R^2 = 0.937}$ in a fully refit temporal walk-forward.

Newscaster — Live since 2024

Agentic AI research and automated publishing system, 525+ episodes — YouTube | GitHub

- Built and operate a daily multi-provider system with adaptive LangGraph research, counter-evidence review, source-grounded fact-checking, TTS, and automated publishing. Newscaster's embedding-based RAG memory ranked the right prior coverage first in **12/12** recurring-story tests, with **93%** of the top six results relevant.

Computational Biology & Genomics

Doctoral Researcher – Computational Neuroscience — 2019–2026

University of California, Riverside — Riverside, CA

- First-author *Journal of Neuroscience* paper: discovered cell-type-specific alternative splicing of *Adgrl2* in single-cell RNA-seq (100K+ cells), then validated it at the bench with mouse genetics, blood-brain-barrier assays, and confocal imaging.
- Built reproducible Python and R pipelines for single-cell and bulk RNA-seq on HPC/SLURM and integrated 6+ public GEO datasets. Trained five researchers, taught for four years, and served as sole instructor for an upper-division course.

Student Researcher – Computational Biology — 2017–2019

Reed College — Portland, OR

- Co-first-author graph-learning study on multi-omic networks (∼65K nodes, ∼7.3M edges), with experimental validation of prioritized disease-gene candidates.

Skills & Education

LANGUAGES & TOOLS	Python, R, SQL, Bash; Git, pytest, HPC/SLURM, Google Cloud Platform, API integration
ML & STATS	scikit-learn, pandas, NumPy; regression, imputation, PLS/PCA/SVD, conformal prediction
AI & EVALUATION	agent failure analysis, benchmark design, LLM-as-judge, RAG/embeddings, LangGraph
BIOINFORMATICS	single-cell/bulk RNA-seq, Seurat, Scanpy, differential expression, alternative splicing, GEO
EDUCATION	Ph.D. Neuroscience, UC Riverside (June 2026); B.S. Neuroscience, Reed College (2019)